QUANTUM MACHINE LEARNING FOR HEALTHCARE

Can a quantum kernel diagnose breast cancer as well as a classical model?

Classical RBF SVM vs quantum ZZ-feature-map kernel SVM on the Wisconsin Diagnostic Breast Cancer dataset

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1 · THE HEALTHCARE PROBLEM

Missing a malignant tumour is the costliest error in diagnosis

Problem

Classify a breast tumour as benign or malignant from biopsy measurements (fine-needle aspirate).

Why it matters

Breast cancer is the most common cancer in women. Early, reliable diagnosis cuts missed cancers and unnecessary procedures.

Intended outcome

High **recall** (few missed cancers) and high ROC-AUC, with a compact model a quantum device can run.

Primary clinical metric: recall (sensitivity) on the malignant class

DATA AND SHARED PIPELINE

Both models see exactly the same data; only the kernel changes

Dataset

Wisconsin Diagnostic Breast Cancer (UCI): 569 patients, 30 features, 212 malignant / 357 benign

Standardise

Z-score scaling, fitted on training folds only (no leakage)

PCA → 6

6 components (88.6% of variance) = 6 qubits

Scale to [0, π]

Angle range for encoding; same features fed to both models

Evaluation: stratified 5-fold outer CV, hyperparameters tuned by inner CV on ROC-AUC, balanced class weights for every model.

Data: Wolberg, Street & Mangasarian, UCI ML Repository (via scikit-learn)

2 · CLASSICAL BASELINE

A strong, comparable baseline: SVM with an RBF kernel

Configuration

- scikit-learn SVC, RBF kernel, class_weight = balanced
- Grid search: C ∈ {0.1, 1, 10, 100}, γ ∈ {scale, 0.01, 0.1, 1}
- Same 6 PCA features as the quantum model
- Extra reference: logistic regression

Why this baseline

The quantum model is also a kernel SVM, so an RBF SVM isolates the effect of the kernel. SVMs are well-established on this dataset (Bennett & Mangasarian; scikit-learn benchmarks).

5-fold CV: AUC 0.996, recall 96.7%

3 · QUANTUM CONTRIBUTION

A quantum kernel replaces the RBF kernel inside the same SVM

How it works

- Encode each patient's 6 features into 6 qubits with a ZZ feature map (Havlíček et al., Nature 2019), 2 repetitions
- Kernel = state overlap: $k(x, x') = |\langle\varphi(x)|\varphi(x')\rangle|^2$
- Same SVM solver uses this kernel (precomputed)
- Bandwidth β scales features before encoding, tuned by inner CV

Why a quantum kernel

- Entangling ZZ gates build pairwise feature interactions into an exponentially large ($2^6 = 64$-dim) feature space
- No trainable circuit parameters, so no barren plateaus and no variational training
- Kernel methods suit small medical datasets

Implementation: exact NumPy statevector simulator, plus a 1,024-shot finite-sampling simulation to mimic hardware

KEY FINDING

The naive quantum kernel failed; tuning its bandwidth fixed it

Naive kernel (β = 1)

0.82 AUC

Recall only 15%: kernel values collapse towards 0 (exponential concentration), so all patients look alike



Tuned bandwidth (β = 0.1)

0.994 AUC

Recall 96.7%: kernel keeps useful similarity; β = 0.1 chosen in all 5 folds

Bandwidth fix follows Shaydulin & Wild, “Importance of kernel bandwidth in quantum machine learning”, Phys. Rev. A 2022

4 · FAIR EVALUATION: FULL DATASET

The quantum kernel matches the classical model on recall

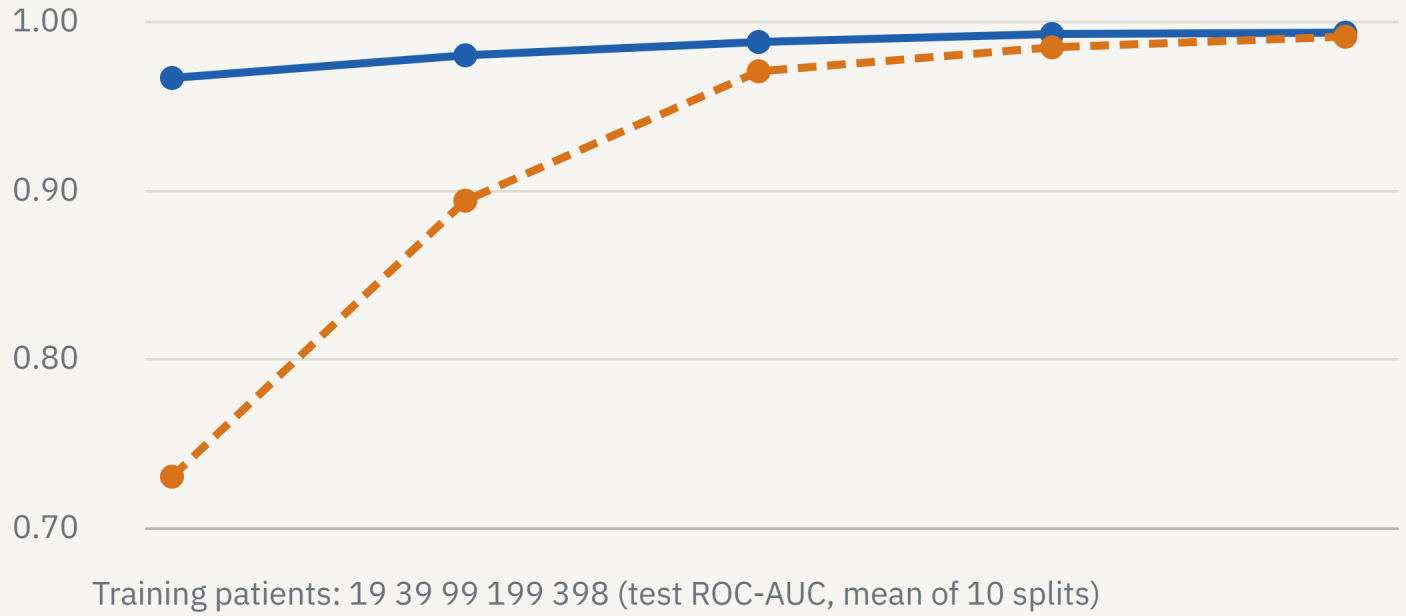
Model (5-fold CV, mean)	Accuracy	Recall	Precision	F1	ROC-AUC
Classical RBF SVM	0.970	0.967	0.955	0.960	0.996
Quantum kernel SVM (exact)	0.963	0.967	0.942	0.952	0.994
Quantum kernel SVM (1,024 shots)	0.953	0.948	0.930	0.937	0.992
Logistic regression (reference)	0.967	0.944	0.967	0.954	0.995

Same recall (96.7%), AUC within 0.002. Precision is about 1.3 points lower. With hardware-like shot noise, AUC stays at 0.992.

Malignant = positive class. Std. dev. across folds: recall ±0.024 for both models; quantum precision ±0.063

SMALL-DATA TEST

With little training data, the classical kernel is still ahead



— Classical RBF SVM

- - Quantum kernel SVM

Gap closes as data grows: 0.24 AUC at 19 patients, 0.003 at 398. Our hoped-for small-data advantage did not appear.

Same fixed 30% test set per split; training subsets stratified; both models tuned by inner CV

QUANTUM RESOURCE AND EFFICIENCY METRICS

A small circuit, but many circuit runs

6
qubits

0
trainable circuit parameters

228
gates per kernel circuit (120 two-qubit)

≤188
circuit depth (upper bound)

Cost item	Classical RBF SVM	Quantum kernel SVM
Kernel entries for full dataset	instant	155,155 circuits
Shots needed (1,024 per circuit)	–	≈159 million
Fit time, 5 folds (simulator)	1.6 s	6.3 s

Kernel circuit = compute–uncompute fidelity test, U(x) then U(x′)†; circuits grow with N^2 training pairs

ADVANTAGES, TRADE-OFFS AND LIMITATIONS

What the quantum kernel offers, and what it costs

Advantages

- Same recall as classical (96.7%): no extra missed cancers
- No trainable circuit parameters: avoids barren plateaus
- Robust to shot noise: AUC 0.992 at 1,024 shots
- Only 6 qubits: fits today's devices

Trade-offs and limitations

- No accuracy advantage; slightly lower precision
- Weaker with very small training sets
- Circuits grow with N^2: costly on real hardware
- Simulated only; real device noise not tested
- PCA to 6 features keeps 88.6% of the information

CONCLUSION

A 6-qubit, parameter-free quantum kernel reaches the same cancer recall as a tuned classical SVM

- Bandwidth tuning is essential: AUC 0.82 → 0.994
- Parity on the full dataset, behind on very small data, robust to shot noise
- Next: run on real quantum hardware, error mitigation, and kernel approximation to cut the N^2 circuit cost

Code: qkernel_breast_cancer.py · results_summary.json